

SynthOCT 2026: Digital Phantoms Simulation for Physics-Based Scans Synthesis in Optical Coherence Tomography: Structured description of the challenge design

CHALLENGE ORGANIZATION

Title

Use the title to convey the essential information on the challenge mission.

SynthOCT 2026: Digital Phantoms Simulation for Physics-Based Scans Synthesis in Optical Coherence Tomography

Challenge acronym

Preferable, provide a short acronym of the challenge (if any).

SynthOCT

Challenge abstract

Provide a summary of the challenge purpose. This should include a general introduction in the topic from both a biomedical as well as from a technical point of view and clearly state the envisioned technical and/or biomedical impact of the challenge.

Optical Coherence Tomography (OCT) is an emerging medical imaging modality and prospective instrument for optical biopsy and virtual histology across various fields such as dermatology, oncology, and beyond. However, training robust foundation models for OCT analysis is hindered by the scarcity of annotated data. While Generative Adversarial Networks (GANs) and diffusion models can synthesize visually plausible images, they often fail to replicate the strict physical laws of OCT scanning and signal formation. These approaches frequently produce "hallucinated" textures that lack physical meaning and consistency. To bridge the gap between synthetic data generation and physical reality, we propose the SynthOCT Challenge.

This challenge focuses on a single, high-quality track: Digital Phantom Generation for Physics-Based Scans Synthesis in Optical Coherence Tomography. Unlike traditional image synthesis tasks, this challenge requires participants to reconstruct the underlying tissue microstructure. Participants will be provided with 120 real OCT B-scans of skin (<https://zenodo.org/records/18095266>) for training and validation. Their task is to generate "Digital Phantoms"—spatial distributions of optical scatterers—that, when virtually scanned, match the target real scans statistically, structurally, and physically. The validity of the synthesis will be assessed using a novel physics-based evaluation framework (https://github.com/SynthOCTChallenge/SynthOCT_Baseline).

We encourage a wide range of open methodologies for creating these scatterer distributions, including but not limited to: deriving structural masks and seeding them with scatterers based on probability distributions; creating dense scatterer masks with parameter optimization; or utilizing end-to-end diffusion models to directly convert scans into scatterer distributions.

Challenge keywords

List the primary keywords that characterize the challenge.

Optical Coherence Tomography, Skin, Synthesis

Year

2026

Novelty of the challenge

Briefly describe the novelty of the challenge.

This challenge represents a paradigm shift from traditional medical image synthesis (which focuses on visual plausibility) to physics-based digital phantom generation. Unlike standard synthesis challenges that rely on perceptual metrics or downstream segmentation performance, SynthOCT evaluates the physical consistency of the generated data. We require participants to reconstruct the underlying tissue microstructure—specifically, the spatial distribution of optical scatterers—rather than just realistic-looking textures.

This is the first challenge to utilize a rigorous evaluation framework based on Optical Attenuation Coefficient (OAC) and Speckle Contrast (SC) maps. By comparing these derived physics-based maps against real scans, we ensure that the synthetic data adheres to the fundamental laws of light-tissue interaction, accurately reflecting the optical properties of the underlying digital phantoms.

The challenge promotes the creation of "Digital Phantoms" that serve as versatile assets for: virtual scanning with adjustable parameters to simulate different device vendors; establishing a foundation for synthetic targeted augmentation of real datasets; generating synthetic scans with perfect ground truth; developing algorithms for translating OCT scans into optical biopsy and virtual histology maps; and creating generalizing datasets with specific feature representations to focus models on critical features during training.

Task description and application scenarios

Briefly describe the application scenarios for the tasks in the challenge.

The organizers provide resources and infrastructure to facilitate participation:

- Cloud-Based Virtual Scanner: Participant-generated phantoms will be processed by our Virtual Scanner—a physics-based simulation engine that replicates the actual optical path, scattering, and interference to render the final synthetic OCT scan. The principles of generation and examples of use utilized in this challenge have been described in recent papers published within the framework of SASHIMI 2024 [1] and 2025 [2] (see Ref. section).

This scanner will be accessible via a robust API, online tools, and standalone software

(https://github.com/SynthOCTChallenge/SynthOCT_Baseline), allowing participants to integrate the rendering engine directly into their development pipelines.

- Code and Baselines: We provide code for generating basic scatterer distributions, examples of digital phantoms, and the necessary evaluation algorithms (including codes for Vermeer's OAC estimation and Speckle Contrast calculation) at the https://github.com/SynthOCTChallenge/SynthOCT_Baseline.

- Data: A training/validation set of 120 real B-scans (<https://zenodo.org/records/18095266>) and a held-out Test set of 60 B-scans for final evaluation (not available during the training stage to avoid leakage).

Evaluation: The challenge focuses on the robustness of the generated digital phantoms. Synthetic scans from participant phantoms will be compared to real target scans using evaluation Multi-Scale Structural Similarity (MS-SSIM) and Learned Perceptual Image Patch Similarity (LPIPS) metrics. MS-SSIM and LPIPS will be applied

across structural and three physics-justified parametric maps to extract evaluation metrics for synthetic-real convergence:

1. Structural Map: Applied to the final B-scan intensity images to assess visual fidelity.
2. Physical Map: Applied to Optical Attenuation Coefficient (OAC) maps, which are critical for tissue characterization.
3. Speckle Statistics: Applied to Speckle Contrast (SC) and Refined Speckle Contrast (RSC) distributions to ensure the synthetic speckle patterns match the physics of real OCT.

Other metrics (such as MSE and PSNR) can be reported only for information due to its failure because of speckles. Please, see the detailed metrics evaluation at:

https://github.com/SynthOCTChallenge/SynthOCT_Baseline/tree/main/Metrics_evaluation

To participate, applicants should register at <http://synthoct.com/>.

In the first stage, applicants upload their digital phantoms for the training set through the application system at <http://synthoct.com/>, where the virtual scanner and evaluation code will be run in the backend; for this stage, we will form a leaderboard.

All participants who achieve structural and physics-based median MS-SSIM metrics higher than a threshold level (Struct > 0.3, OAC>0.4, SC>0.4 and RSC>0.5) and LPIPS<0.4 (for all four maps) are invited to upload their code and a full paper for the final evaluation. Their papers should be presented at the SynthOCT Challenge held jointly with the SASHIMI 2026 Workshop at MICCAI 2026.

During the paper review process, their code and models will be validated by the organizers on the final test dataset (60 B-scans), which is not available during the first stage. The top three results based on SSIM metrics on the test set will be the winner and two runners-up. The final results will be announced on-site at MICCAI 2026 (SynthOCT/SASHIMI 2026).

Further applications of the algorithms resulting from the Challenge include, but are not limited to:

Generating diverse, physically accurate training data with specific feature representations to improve the robustness of foundation models (e.g., MedSAM) for segmentation and classification in dermatology.

Creating "Digital Phantoms" to test and validate OCT hardware, enabling the simulation of devices from different vendors and the in silico evaluation of algorithms under varying and controllable physical parameters.

Distributing synthetic datasets that retain the statistical and physical properties of real patient data without compromising patient privacy.

Developing virtual histology maps based on scatterer distributions.

FURTHER INFORMATION FOR CONFERENCE ORGANIZERS

Workshop

If the challenge is part of a workshop, please indicate the workshop.

SASHIMI Workshop. Samuel Remedios, the SASHIMI co-organizer, is also a co-organizer of this challenge proposal. We have already discussed and agreed upon this merge with the SASHIMI organizers

Duration

How long does the challenge take?

2 Hours

In case you selected half or full day, please explain why you need a long slot for your challenge.

This challenge is proposed to be held as a satellite event merged with the SASHIMI 2026 Workshop. We have already discussed an integration with the SASHIMI organizers. The SASHIMI 2026 organizers have submitted their own proposal and have also indicated a merge with SynthOCT. Both SynthOCT and SASHIMI plan to occupy the same half-day timeslot; therefore, we indicated the duration above as 2 hours.

Expected number of participants

Please explain the basis of your estimate (e.g. numbers from previous challenges) and/or provide a list of potential participants and indicate if they have already confirmed their willingness to contribute.

Expected number of participants 20-30 teams.

Basis of estimate: This estimate is based on the active community established during the related IEEE VIP Cup 2024, which was devoted to OCT signal reconstruction. We are targeting research groups with experience in generative AI for medical imaging from the following institutions:

Lobachevsky State University of Nizhny Novgorod (Participation confirmed)

HSE University (Participation confirmed)

Isfahan University of Technology

Indian Institute of Technology Ropar

Vietnamese-German University

Khalifa University

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)

Saratov State University

Lomonosov Moscow State University

Samara National Research University

National University of Sciences & Technology (NUST), Pakistan

University of Toronto

University of Western Australia

University College London (UCL)

University of Rochester

Aston University

Abu Dhabi University

New York University (NYU)

University of Basel

Victoria University of Wellington

Toronto Metropolitan University

And other research centers developing GAN/diffusion models for OCT.

Publication and future plans

Please indicate if you plan to coordinate a publication of the challenge results.

We plan to invite participants to publish their approaches within the joint SynthOCT and SASHIMI 2026 proceedings. Publication is contingent upon the provision of open-source code for the final evaluation phase to ensure reproducibility.

Publication in the MICCAI LNCS series is planned for this Challenge. This serves not only as motivation for the participants but also as a means of making their approaches, code, and digital phantoms publicly available.

MICCAI LNCS proceedings

Indicate if you want to offer MICCAI Springer LNCS proceedings to the participants. Publishing a proceedings volume is optional and at the discretion of each challenge's organizers. At a minimum, organizers must ensure that a description of each participant's submission is publicly available. Organizers who wish to publish MICCAI Springer LNCS proceedings must adhere to the MICCAI Satellite events publication process.

Yes

Collaboration with European Society of Radiology (ESR)

In collaboration with European Society of Radiology (ESR), we announce special clinical interest topics with associated clinicians who can help with the preparation of the proposals; the best 3 challenge proposals on these topics will get the opportunity to present their challenges at the European Congress of Radiology (ECR) 2027 in a special session. If you want to organize a challenge in collaboration with ESR on one of these topics, please reach out to the MICCAI Challenges Team (miccai-challenges-2026@dkfz-heidelberg.de) and we will put you in contact with the corresponding clinician.

Challenge in collaboration with ESR. Ticking 'Yes' implies that the challenge has been prepared in collaboration with the clinical contact point.

No

Space and hardware requirements

Organizers of on-site challenges must provide a fair computing environment for all participants. For instance, algorithms should run on the same computing platform provided to all.

We plan to evaluate the participants' approaches in advance during the paper and code review process. For on-site facilities, we foresee the need for space and standard hardware (projector, screen, computer) for the presentation of their reports.

TASK 1: Digital Phantom Generation for Physics-Based Scans Synthesis in Optical Coherence Tomography

SUMMARY

Abstract

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Optical Coherence Tomography (OCT) is an emerging medical imaging modality and prospective instrument for optical biopsy and virtual histology across various fields such as dermatology, oncology, and beyond. However, training robust foundation models for OCT analysis is hindered by the scarcity of annotated data. While Generative Adversarial Networks (GANs) and diffusion models can synthesize visually plausible images, they often fail to replicate the strict physical laws of OCT scanning and signal formation. These approaches frequently produce "hallucinated" textures that lack physical meaning and consistency. To bridge the gap between synthetic data generation and physical reality, we propose the SynthOCT Challenge.

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Keywords

List the primary keywords that characterize the task.

Optical Coherence Tomography, Skin, Synthesis

ORGANIZATION

Organizers

a) Provide information on the organizing team (names and affiliations).

The organizing team combines extensive expertise in synthetic data generation, physics-based modeling, OCT signal formation, and biomedical image processing. Furthermore, the inclusion of local organizers from major universities in the UAE is intended to foster strong regional participation and attract local academic teams to the tournament.

Lev Matveev (Russian Academy of Sciences) Email: drlevmatveev@gmail.com

Samuel Remedios (Johns Hopkins University) Email: samuel.remedios@jhu.edu

Azhar Zam (New York University / NYU Abu Dhabi) Email: azhar.zam@nyu.edu

Taimur Hassan (Abu Dhabi University) Email: taimur.hassan@adu.ac.ae

Notes:

Prof. Azhar Zam was also an organizer of the 2024 IEEE VIP Cup devoted to OCT scans reconstruction (<https://2024.ieeeicip.org/vip-cup/>)

Dr. Lev Matveev is the MICCAI member as required by application rules (User ID 77440252)

b) Provide information on the primary contact person.

Dr . Lev Matveev (Russian Academy of Sciences), MICCAI Member (User ID 77440252)

Email: lev@ipfran.ru; drlevmatveev@gmail.com

c) Indicate whether clinicians are part of the organizing team. If yes, describe their role.

Our task is based on the dataset available on Zenodo: <https://zenodo.org/records/18095266> . No clinicians are required as part of the organizing team.

Life cycle type

Define the intended submission cycle of the challenge. Include information on whether/how the challenge will be continued after the challenge has taken place. Not every challenge closes after the submission deadline (one-time event). Sometimes it is possible to submit results after the deadline (open call) or the challenge is repeated with some modifications (repeated event).

Examples:

- One-time event with fixed conference submission deadline

- Open call (challenge opens for new submissions after conference deadline)
- Repeated event with annual fixed conference submission deadline

One-time event with fixed conference submission deadline

Challenge venue and platform

a) Report the event (e.g. conference) that is associated with the challenge (if any).

MICCAI 2026 / SASHIMI 2026 Workshop

b) Report the platform used to run the challenge.

We will host our own platform at <http://synthoct.com/>

c) Do you agree that the your submission is shared with the platform (e.g., grand-challenge, synapse...) that you indicated?

Please note: 1) this purpose of such sharing is that the challenge chairs and the platform can communicate smoothly, your answer won't impact the review of your proposal; 2) regardless of your response to this question, it is your responsibility to perform all actions required by the platform (e.g. filling their submission request).

Yes

d) Provide the URL for the challenge website (if any).

<http://synthoct.com/>

Participation policies

a) Define the allowed user interaction of the algorithms assessed. This includes the policy regarding any curation, (pre-)processing and (pre-)training steps.

Semi Automatic; The dataset is available on Zenodo. The registration and submission system will be provided. The evaluation of digital phantoms in the first stage will be fully automated. The final test will be performed by the organizers.

b) Define the policy on the usage of training data. The data used to train algorithms may, for example, be restricted to the data provided by the challenge or may also include publicly available data including (open) pre-trained nets. Clarify whether such additional data needs to be publicly available at the time of the challenge launch. Clarify whether adding (private) annotations of the public data is allowed.

No policy defined; Participants may use any pre-trained models and algorithms; adding external private or public data is allowed.

c) Define the participation policy for members of the organizers' institutes. For example, members of the organizers' institutes may participate in the challenge but are not eligible for awards.

To promote inclusivity and ensure a fair competition, participants from the host institutions must not have any personal or professional ties to the organizing committee members or their respective laboratories; however, students from the general student body of these universities who are not directly linked to the organizers are eligible to participate.

d) Define the award policy. In particular, provide details with respect to challenge prizes.

Certificates: Official certificates will be awarded to the top-3 performing teams.

Publication: Participants are invited to submit papers describing their approaches to the MICCAI Satellite Events proceedings (Springer LNCS). Additionally, members of the top-performing teams (meeting the MS-SSIM > threshold and LPIPS < threshold) will be invited to co-author the joint challenge summary paper.

Sponsorship: The organizers are currently conducting fundraising activities to secure monetary prizes for the winners. Any secured prize pool will be announced to participants prior to the final test phase.

e) Define the policy for result announcement.

Examples:

- Top 3 performing methods will be announced publicly.
- Participating teams can choose whether the performance results will be made public.

Top 3 performing methods will be announced publicly

f) Define the publication policy. In particular, provide details on ...

- ... who of the participating teams/the participating teams' members qualifies as author
- ... whether the participating teams may publish their own results separately, and (if so)
- ... whether an embargo time is defined (so that challenge organizers can publish a challenge paper first).

Authorship: Members of the top participating teams (specifically those achieving median metrics above the threshold) qualify as co-authors on the challenge summary paper.

Separate Publication: Participating teams may publish their own results separately within the challenge proceedings.

Embargo: Teams are allowed to publish follow-up papers developing their approaches after the MICCAI 2026 conference.

Submission method

a) Describe the method used for result submission. Preferably, provide a link to the submission instructions.

Examples:

- Docker container on the Synapse platform. Link to submission instructions: <URL>
- Algorithm output was sent to organizers via e-mail. Submission instructions were sent by e-mail.

Stage 1: Algorithm outputs are submitted to the organizers via the challenge website and email.

Stage 2: Algorithm code and models must be made open-access. A link to the repository must be provided together with the paper submission before the deadline. The repository must be accessible for review and made public after the proceedings are published.

b) Provide information on the possibility for participating teams to evaluate their algorithms before submitting final results. For example, many challenges allow submission of multiple results, and only the last run is officially counted to compute challenge results.

The evaluation codes are publicly available at the Challenge Repo

(https://github.com/SynthOCTChallenge/SynthOCT_Baseline), so at the first stage we recommend validating the results before submitting digital phantoms. To avoid overloading the online submission system, we allow only 3

online submissions of full results for each participant (last submission is officially counted to compute challenge results).

Challenge schedule

Provide a timetable for the challenge. Preferably, this should include

- the release date(s) of the training cases (if any)
- the registration date/period
- the release date(s) of the test cases and validation cases (if any)
- the submission date(s)
- associated workshop days (if any)
- the release date(s) of the results

Training and validation datasets: Already released (<https://zenodo.org/records/18095266>)

Registration period: Within two months of Challenge acceptance

First stage submission deadline: Before June 2026

Full paper and code submission: Before July 2026

Full paper acceptance notification: Before August 2026

SynthOCT Challenge (jointly with Sashimi Workshop): October 4 or 8, 2026

Results announcement: After the final presentations on the day of the Challenge

Ethics approval

Indicate whether ethics approval is necessary for the data. If yes, provide details on the ethics approval, preferably institutional review board, location, date and number of the ethics approval (if applicable). Add the URL or a reference to the document of the ethics approval (if available).

The data does not require any additional approvals because the dataset was gathered under the legal procedures described in the paper: A. A. Sovetsky, K. S. Petrova, M. A. Brueva, M. G. Ryabkov, A. L. Matveyev, L. A. Matveev, V. Y. Zaitsev. "Automated segmentation and skin-layer thickness estimation by extracting the optical scattering coefficient and speckle contrast parameter from optical coherence tomography scans". Skin Pharmacology and Physiology, 2026 (DOI: 10.1159/000550613)

Data usage agreement

Clarify how the data can be used and distributed by the teams that participate in the challenge and by others during and after the challenge. This should include the explicit listing of the license applied.

Examples:

- CC BY (Attribution)
- CC BY-SA (Attribution-ShareAlike)
- CC BY-ND (Attribution-NoDerivs)
- CC BY-NC (Attribution-NonCommercial)
- CC BY-NC-SA (Attribution-NonCommercial-ShareAlike)

- CC BY-NC-ND (Attribution-NonCommercial-NoDerivs)

Please note that the data license should not differ among sources. In case a license has to be changed, it has to be reported to the MICCAI challenges team and changed in the proposal.

CC BY (Attribution)

Code availability

a) Provide information on the accessibility of the organizers' evaluation software (e.g. code to produce rankings). Preferably, provide a link to the code and add information on the supported platforms.

Codes and software: https://github.com/SynthOCTChallenge/SynthOCT_Baseline

Online platform: <https://accounts.opticelastograph.com>

Challenge platform: <http://synthoct.com>

b) In an analogous manner, provide information on the accessibility of the participating teams' code.

Digital phantom generation code should be provided in the participants' repositories, e.g. on GitHub

Conflicts of interest

Provide information related to conflicts of interest. In particular provide information related to sponsoring/funding of the challenge. Also, state explicitly who had/will have access to the test case labels and when.

This Challenge is partially supported by the Russian Science Foundation (RSF) Grant № 25-12-20032: "New Approaches to the Development of Algorithms for Analyzing OCT Scans: Modification and Optimization of Large Models Based on Physical Principles and Conditions of OCT Signal Formation"

(<https://rscf.ru/en/project/25-12-20032/>)

The final test dataset is with Dr. Lev Matveev and co-authors of the paper A. A. Sovetsky, K. S. Petrova, M. A. Brueva, M. G. Ryabkov, A. L. Matveyev, L. A. Matveev, V. Y. Zaitsev. "Automated segmentation and skin-layer thickness estimation by extracting the optical scattering coefficient and speckle contrast parameter from optical coherence tomography scans". Skin Pharmacology and Physiology, 2026

We confirm that we use the CC BY 4.0 license. The data is hosted on Zenodo providing open access without registration barriers, which is fully consistent with the license terms and the guidelines provided by The MICCAI 2026 Challenge Team.

Note: The training dataset of 120 B-scans (in-sample) is freely available at zenodo.org. The test dataset (out-of-sample) consisting of 60 B-scans will be added there after MICCAI 2026 to avoid test data leakage.

MISSION OF THE CHALLENGE

Field(s) of application

State the main field(s) of application that the participating algorithms target.

Examples:

- Diagnosis
- Education
- Intervention assistance

- Intervention follow-up
- Intervention planning
- Prognosis
- Research
- Screening
- Training
- Cross-phase

Research

Task category(ies)

State the task category(ies)

Examples:

- Classification
- Detection
- Localization
- Modeling
- Prediction
- Reconstruction
- Registration
- Retrieval
- Segmentation
- Tracking

Simulation

Cohorts

We distinguish between the target cohort and the challenge cohort. For example, a challenge could be designed around the task of medical instrument tracking in robotic kidney surgery. While the challenge could be based on ex vivo data obtained from a laparoscopic training environment with porcine organs (challenge cohort), the final biomedical application (i.e. robotic kidney surgery) would be targeted on real patients with certain characteristics defined by inclusion criteria such as restrictions regarding sex or age (target cohort).

a) Describe the target cohort, i.e. the subjects/objects from whom/which the data would be acquired in the final biomedical application.

The target cohort consists of patients undergoing dermatological screening or diagnosis. Consequently, the object of interest is human skin tissue imaged via Optical Coherence Tomography (OCT). The participating algorithms target the generation of digital phantoms from these skin OCT scans to enhance data augmentation,

generalization approaches, and virtual biopsy (virtual histology) techniques. Furthermore, the scope of this research extends beyond dermatology, opening opportunities to transfer these algorithms to other OCT-based tasks, such as oncology.

b) Describe the challenge cohort, i.e. the subject(s)/object(s) from whom/which the challenge data was acquired.

The data was acquired from volunteers for the research proposal. Details are provided in A. A. Sovetsky, K. S. Petrova, M. A. Brueva, M. G. Ryabkov, A. L. Matveyev, L. A. Matveev, V. Y. Zaitsev. "Automated segmentation and skin-layer thickness estimation by extracting the optical scattering coefficient and speckle contrast parameter from optical coherence tomography scans". Skin Pharmacology and Physiology, 2026 and Petrova GA, Petrova KS, Nemirova SV, Karpenko AA. 3D optical coherence tomography: possibilities in the diagnosing skin neoplasms. Russian Journal of Clinical Dermatology and Venereology. 2019;18(6):715727. (In Russ.) <https://doi.org/10.17116/klinderma201918061715>

Imaging modality(ies)

Specify the imaging technique(s) applied in the challenge.

Optical coherence tomography

Context information

Provide additional information given along with the images. The information may correspond ...

a) ... directly to the image data (e.g. tumor volume).

Each skin OCT B-scan is provided in two formats - .npy (NumPy) and .png - Visual representation of the B-scans for quick inspection.

Raw data containing the logarithmic signal amplitude.

Dimensions: 256 (depth) x 512 (lateral) pixels

Type: uint8 (0-255)

Scale: Logarithmic (~51 dB dynamic range, where 1 unit $\approx$ 0.2 dB).

Pixel Size: 6 μ m (axial) x 6 μ m (lateral)

FWHM Beam Diameter: ~20 μ m

Center Wavelength: 1.3 μ m

b) ... to the patient in general (e.g. sex, medical history).

Data Structure and Format The dataset is organized into a folder hierarchy:

Sex (Male, Female) / Age Group (1950-1960, 1990-2000) / Localization (Cheek, Eye_corner).

Target entity(ies)

a) Describe the data origin, i.e. the region(s)/part(s) of subject(s)/object(s) from whom/which the image data would be acquired in the final biomedical application (e.g. brain shown in computed tomography (CT) data, abdomen shown in laparoscopic video data, operating room shown in video data, thorax shown in fluoroscopy video). If necessary, differentiate between target and challenge cohort.

The work was conducted between 2019 and 2025 at the Department of Dermatology and Venereology, Privolzhsky Research Medical University (PRMU), Ministry of Health of the Russian Federation.

b) Describe the algorithm target, i.e. the structure(s)/subject(s)/object(s)/component(s) that the participating algorithms have been designed to focus on (e.g. tumor in the brain, tip of a medical instrument, nurse in an operating

theater, catheter in a fluoroscopy scan). If necessary, differentiate between target and challenge cohort.

The participating algorithms focus on the microstructural optical properties of biological tissue, as well as its meso- and macro-architecture, based on the distribution of optical scatterers within skin layers and inclusions. The main targets of the algorithms are:

Scatterer Distribution Reconstruction: Algorithms must accurately infer and generate scatterer distributions based on input OCT B-scans (e.g., defining the tissue architecture and seeding regions with scatterers).

Digital Phantom Generation: The creation of "OCT-ready" digital phantoms that serve as high-fidelity synthetic data for training downstream analysis models (e.g., segmentation networks).

Virtual Histology Mapping (potential for clinical scaling): Solving the inverse problem—the creation of scatterer-based maps from real scans—allows for mimicking histological staining, enabling non-invasive 'virtual biopsy' capabilities.

Assessment aim(s)

Identify the property(ies) of the algorithms to be optimized to perform well in the challenge. If multiple properties are assessed, prioritize them (if appropriate). The properties should then be reflected in the metrics applied (see below, parameter metric(s)), and the priorities should be reflected in the ranking when combining multiple metrics that assess different properties.

- Example 1: Find highly accurate liver segmentation algorithm for CT images.
- Example 2: Find lung tumor detection algorithm with high sensitivity and specificity for mammography images.

Corresponding metrics are listed below (parameter metric(s)).

Fidelity, Reconstruction Accuracy

DATA SETS

Data source(s)

a) Specify the device(s) used to acquire the challenge data. This includes details on the device(s) used to acquire the imaging data (e.g. manufacturer) as well as information on additional devices used for performance assessment (e.g. tracking system used in a surgical setting).

OCT setup (designed and produced in the Institute of Applied Physics RAS) with the central wavelength 1.3 micrometers and spectral width 90 nm. The acquisition rate of the setup was 92 kA-scan per second. The radiation power on the object is 0.75 - 6 - mW (below the level allowed by the AMSI standard), the scanning depth was up to 2 mm in air, each A-scan was composed of 256 pixels, and B-scan width was 512 px covering 5 mm laterally in the x- direction of fast scanning. The probe also enables scanning in y-direction allowing for obtaining 3D data sets covering the scanning area 5x3 mm (512x512 pixels in both directions). The device was equipped with a removable flexible probe with the outer diameter of the scanning head 10 mm.

b) Describe relevant details on the imaging process/data acquisition for each acquisition device (e.g. image acquisition protocol(s)).

The in vivo OCT data on patients' healthy skin were collected from patients of both sexes and different ages (born in the period from ~1950 to ~2000 years divided in six 10-year groups), comprising 38 women and 12 men. For each person, 8 localization on faces were examined (two localizations in the centers of forehead and chin, two for left/right cheeks, two for eye corners and two close places around the axis of the neck). After eliminating a small

portion of low-quality records, totally 282 sets of 3D OCT data each containing 512 B-scans were analyzed for women and 96 similar 3D data sets for men. Voluntary informed consent of the patients for the OCT examination and subsequent analysis of OCT data was obtained.

c) Specify the center(s)/institute(s) in which the data was acquired and/or the data providing platform/source (e.g. previous challenge). If this information is not provided (e.g. for anonymization reasons), specify why.

Privolzhsky Research Medical University (PRMU), Ministry of Health of the Russian Federation

d) Describe relevant characteristics (e.g. level of expertise) of the subjects (e.g. surgeon)/objects (e.g. robot) involved in the data acquisition process (if any).

The dataset was acquired by clinicians and dermatologists: K. S. Petrova, M. A. Brueva, and M. G. Ryabkov.

Training and test case characteristics

a) State what is meant by one case in this challenge. A case encompasses all data that is processed to produce one result that is compared to the corresponding reference result (i.e. the desired algorithm output).

Examples:

- Training and test cases both represent a CT image of a human brain. Training cases have a weak annotation (tumor present or not and tumor volume (if any)) while the test cases are annotated with the tumor contour (if any).
- A case refers to all information that is available for one particular patient in a specific study. This information always includes the image information as specified in data source(s) (see above) and may include context information (see above). Both training and test cases are annotated with survival (binary) 5 years after (first) image was taken.

A case is defined as a single in vivo Optical Coherence Tomography (OCT) B-scan of human skin. The dataset consists of 120+60 such scans acquired from volunteers. For the challenge, these real in vivo scans serve as the reference (ground truth). The participating algorithms must generate synthetic "digital phantoms" that statistically and structurally (with physics-based parametric maps) match these real cases after virtual scanning. Specifically, the generated digital phantom (scatterer distribution) undergoes a virtual scanning simulation to produce a synthetic OCT B-scan, which is then compared to the original real in vivo scan to assess realism and physics consistency.

b) State the total number of training, validation and test cases.

120 for training and validation + 60 for tests.

Since the out-of-sample final test set (60 B-scans) is being held back to avoid leakage, participants are free to split the 120 B-scan dataset into training and validation sets as they see fit to prevent overfitting. However, improper data handling or partitioning remains the participants' responsibility and may negatively impact their final test results.

c) How much of the data are already annotated (stratified by train test in percentage)?

No annotation is required

d) Explain why a total number of cases and the specific proportion of training, validation and test cases was chosen.

The full source dataset comprises 282 3D OCT volumes from female subjects and 96 volumes from male subjects, totaling 193,536 B-scans.

For this challenge, we have selected a curated subset of 120 B-scans for the development phase (Training/Validation) and an additional 60 B-scans reserved for the final hidden test phase.

The rationale for this selection is based on two key factors:

Computational Feasibility: The challenge requires participants to generate digital phantoms and perform virtual scanning simulations to validate them against real scans. This process is computationally intensive. Limiting the dataset to 120 total cases ensures the task remains feasible for participants' hardware and allows organizers to conduct the final evaluation within a reasonable timeframe.

Sufficiency for Generalization: The algorithms are expected to ensure convergence and high fidelity for each B-scan individually (solving the inverse problem per case). Since the selected cases are stratified by biological sex, age, and anatomical localization, this subset of 120+60 scans covers the necessary biological variance.

Successfully processing this diverse subset is sufficient to demonstrate the method's generalization capabilities without the computational overhead of a larger dataset.

e) Mention further important characteristics of the training, validation and test cases (e.g. class distribution in classification tasks chosen according to real-world distribution vs. equal class distribution) and justify the choice.

Although the primary challenge task is data generation rather than classification, maintaining a balanced class distribution is crucial to ensure the algorithms generalize across different biological phenotypes. Therefore, the training and validation dataset (120 B-scans) is perfectly stratified across three binary categories: Biological Sex (Male/Female), Age Group (1950-1960 vs. 1990-2000), and Anatomical Localization (Cheek vs. Eye Corner). This structure is visible in the dataset repository: <https://zenodo.org/records/18095266>. The final test set (60 B-scans) follows this identical stratified distribution to ensure a fair and comprehensive evaluation of model robustness across all subgroups.

f) Challenge organizers are encouraged to (partly) use unseen, unpublished data for their challenges. Describe if new data will be used for the challenge and state the number of cases along with the proportion of new data.

The training and validation dataset (120 cases) is already published and available on Zenodo (<https://zenodo.org/records/18095266>).

For the final evaluation, we will use a set of 60 unpublished cases. These cases originate from the same cohort and follow the same acquisition protocol as the public data but have been withheld to ensure a fair evaluation and prevent final test data leakage.

Annotation characteristics

a) Describe the method for determining the reference annotation, i.e. the desired algorithm output. Provide the information separately for the training, validation and test cases if necessary. Possible methods include manual image annotation, in silico ground truth generation and annotation by automatic methods.

If human annotation was involved, state the number of annotators.

No manual annotation is required. The reference standard (ground truth) is the original in vivo OCT B-scan itself. The participating algorithms must reconstruct the scatterer distribution within a digital phantom to ensure that the resulting synthetic scan converges with the real reference scan in terms of structural and statistical similarity.

b) Provide the instructions given to the annotators (if any) prior to the annotation. This may include description of a training phase with the software. Provide the information separately for the training, validation and test cases if

necessary. Preferably, provide a link to the annotation protocol.

Not applicable

c) Provide details on the subject(s)/algorithm(s) that annotated the cases (e.g. information on level of expertise such as number of years of professional experience, medically-trained or not). Provide the information separately for the training, validation and test cases if necessary.

Not applicable

d) Describe the method(s) used to merge multiple annotations for one case (if any). Provide the information separately for the training, validation and test cases if necessary.

Not applicable

Data pre-processing method(s)

Describe the method(s) used for pre-processing the raw training data before it is provided to the participating teams. Provide the information separately for the training, validation and test cases if necessary.

No specific pre-processing was performed. The data is provided in its raw format (logarithmic signal amplitude)

Sources of error

a) Describe the most relevant possible error sources related to the image annotation. If possible, estimate the magnitude (range) of these errors, using inter-and intra-annotator variability, for example. Provide the information separately for the training, validation and test cases, if necessary.

Not applicable

b) In an analogous manner, describe and quantify other relevant sources of error.

A primary source of uncertainty stems from the physical resolution limit of standard OCT systems, which is insufficient to resolve individual optical scatterers directly. Consequently, establishing a direct ground truth at the single-scatterer level is not feasible. Instead, validation relies on indirect parametric maps—specifically the Optical Attenuation Coefficient (OAC), Speckle Contrast (SC), and Refined Speckle Contrast (RSC)—which are analytically linked to the statistical parameters of the scatterer distribution. However, a significant source of error in this approach is the high sensitivity of OAC estimation to the Signal-to-Noise Ratio (SNR), device parameters and to artifacts arising from noise addition or suppression during the scan generation process (for more info see e.g. <https://doi.org/10.1364/BOE.5.000322> , <https://doi.org/10.1088/1612-202X/acd7e2> , [10.18287/JBPE25.11.030301](https://doi.org/10.18287/JBPE25.11.030301) , [10.18287/JBPE24.10.020302](https://doi.org/10.18287/JBPE24.10.020302)). To avoid this source of uncertainty, we are providing a set of OCT device parameters that should be integrated into the models. Insufficient accuracy in the central wavelength, beam radius, or axial resolution will lead to shifts in speckle-governing distributions at the microstructural level. These factors are taken into account during the metrics evaluation (https://github.com/SynthOCTChallenge/SynthOCT_Baseline/tree/main/Metrics_evaluation)

ASSESSMENT METHODS

Metric(s)

a) Define the metric(s) to assess a property of an algorithm. These metrics should reflect the desired algorithm properties described in assessment aim(s) (see above). State which metric(s) were used to compute the ranking(s) (if any).

- Example 1: Dice Similarity Coefficient (DSC)
- Example 2: Area under curve (AUC)

The challenge focuses on the consistency of the generated digital phantoms. Synthetic scans from participant phantoms will be compared to real target scans using evaluation Multi-Scale Structural Similarity (MS-SSIM) and Learned Perceptual Image Patch Similarity (LPIPS) metrics (MS-SSIM from sewar Python library with default MS-SSIM parameters will be used and LPIPS from Lpips library in Python). MS-SSIM and LPIPS will be applied across structural and three physics-justified parametric maps to extract evaluation metrics for synthetic-real convergence:

1. Structural Map: Applied to the final B-scan intensity images to assess visual fidelity.
2. Physical Map: Applied to Optical Attenuation Coefficient (OAC) maps, which are critical for tissue characterization.
3. Speckle Statistics: Applied to Speckle Contrast (SC) and Refined Speckle Contrast (RSC) distributions to ensure the synthetic speckle patterns match the physics of real OCT.

b) Justify why the metric(s) was/were chosen, preferably with reference to the biomedical application.

We implement the metrics evaluation by aggregating performance statistics for OCT-specific scans and physics-based maps in our repository:

https://github.com/SynthOCTChallenge/SynthOCT_Baseline/tree/main/Metrics_evaluation

The Multi-Scale Structural Similarity Index Measure (MS-SSIM) and Learned Perceptual Image Patch Similarity (LPIPS) were selected as the primary metrics because it assesses structural information and texture preservation better than pixel-wise metrics (e.g., MSE, PSNR). However, for this challenge, visual similarity of the intensity B-scans is necessary but not sufficient. We extend the evaluation to physics-based parametric maps to ensure the generated digital phantoms are physically accurate, not just visually plausible. The justification for this multi-parametric approach is as follows:

Speckle Contrast (SC) and RSC: These metrics are highly sensitive to the statistical distribution of scatterers within the resolution elements (coherence volume). High MS-SSIM on these maps (as illustrated in Metrics_evaluation repo) confirm that the algorithm has correctly modeled the sub-resolution arrangement of scatterers.

Optical Attenuation Coefficient (OAC): This parameter is sensitive to the backscattering strength and the local concentration of scatterers. Low LPIPS scores, coupled with high MS-SSIM values on OAC maps, demonstrate the correctness of the amplitude parameterization and the spatial distribution (seeding density) of scatterers across different tissue regions.

Thus, achieving high MS-SSIM and Low LPIPS across all four maps (Raw OCT, OAC, SC, RSC) ensures that the inverse problem of reconstructing the tissue's physical microstructure has been solved correctly.

Ranking method(s)

a) Describe the method used to compute a performance rank for all submitted algorithms based on the generated metric results on the test cases. Typically the text will describe how results obtained per case and metric are aggregated to arrive at a final score/ranking. Ideally, provide the ranking scheme as a concrete pseudo code.

The final ranking (Main Leaderboard) will be determined by the arithmetic mean of the median MS-SSIMs and inverted LPIPS ($1 - \text{LPIPS}$) values, calculated across four evaluation categories: Structural Intensity, Optical Attenuation Coefficient (OAC), Speckle Contrast (SC), and Refined Speckle Contrast (RSC).

Ranking Logic:

Per-Metric Aggregation: For each of the four metrics, we compute the median SSIM score across all test cases. The median is chosen to ensure robustness against outliers in individual scans.

Global Score: The final score is the average of these four median values.

Pseudo-code:

```
def compute_rank(participant_results):
    # participant_results: list of dicts for each test case
    # metrics = ["MSSSIM_Struct", "MSSSIM_OAC", "MSSSIM_SC", "1-LPIPS_RSC", "1-LPIPS_Struct", "1-LPIPS_OAC",
    "1-LPIPS_SC", "1-LPIPS_RSC"]

    metric_medians = {}

    # Step 1: Calculate Median for each metric across all test cases
    for metric in metrics:
        all_scores_for_metric = [case[metric] for case in participant_results]
        metric_medians[metric] = median(all_scores_for_metric)

    # Step 2: Compute Global Score (Average of the 8 medians)
    global_score = mean(metric_medians.values())

    return global_score, metric_medians
```

Detailed Analysis of Trade-offs: We anticipate that a single global score may mask specific algorithmic behaviors. For instance, an algorithm might achieve high MS-SSIM for Structural and RSC maps but show a performance drop in OAC estimation. Conversely, another participant might demonstrate superior OAC convergence due to better noise modeling and OAC SNR optimization (as noted in "Other sources of error"), despite slightly lower structural fidelity. Therefore, in addition to the main leaderboard, we will report sub-rankings for each metric group. We will explicitly analyze and highlight these inter-metric trade-offs in the final results to provide a comprehensive view of each method's strengths (e.g., "Visually Faithful" vs. "Physically Accurate").

b) Describe the method(s) used to manage submissions with missing results on test cases.

We enforce a uniform execution pipeline where all 60 test B-scans are processed sequentially by the participants' algorithms. Given this structure, we anticipate that algorithms will either successfully process the entire batch or fail to execute completely due to systemic errors (e.g., code crashes). Therefore, partial missing results are not expected.

In the event of technical failures (e.g., runtime errors during the execution), we will contact the corresponding authors to resolve these specific technical issues to ensure a valid submission.

c) Justify why the described ranking scheme(s) was/were used.

The aggregation of results using the median MS-SSIM and 1-LPIPS across all test cases was chosen for its robustness and simplicity. Unlike the mean, the median is less sensitive to outliers. This ensures that a participant's ranking is determined by their algorithm's typical performance in generating scatterer distributions, rather than being disproportionately penalized by a single failed case (e.g., an artifact in one specific B-scan). Furthermore, the calculation of confidence intervals (e.g., via bootstrapping) is crucial for the final analysis. It

allows us to determine whether the performance differences between top-ranked algorithms are statistically significant or merely due to random variance, ensuring a fair and scientifically valid declaration of the winners.

Statistical analyses

Provide an overview of the statistical approaches used in the scope of the challenge analysis. Details can be provided in the parameters below. For each parameter, justify why the described statistical method(s) was/were used and, if necessary, add a description of any method used to assess whether the data met the assumptions required for the particular statistical approach.

The statistical analysis framework is designed to evaluate both the accuracy and the robustness of the submitted algorithms. The approach includes:

Assessment of Variability: We will analyze the distribution of metric values (MS-SSIM and LPIPS) across the 60 test cases (variance between scans).

Confidence Intervals: To determine the reliability of the ranking, we will construct 95% Confidence Intervals (CIs) for the final aggregate scores using a non-parametric bootstrap approach (sampling with replacement).

Comparative Analysis: We will compare these confidence intervals between participants.

Provide a description of how the precision of the performance estimates of individual algorithms is assessed (e.g. confidence interval of the mean on the test set computed using percentile bootstrap, confidence interval of the accuracy on the test set computed using percentile bootstrap).

The precision of the performance estimates will be assessed by computing 95% Confidence Intervals (CIs) for the median SSIM values.

We interpret these estimates within the context of the physical limits of the imaging modality. Specifically:

Upper Bound: We anticipate that the maximum achievable SSIM is bounded by the intrinsic Signal-to-Noise Ratio (SNR) of the real in vivo scans and the precision of the device parameterization, rather than reaching a theoretical 1.0.

Performance Baselines: Based on our preliminary tests (available at:

https://github.com/SynthOCTChallenge/SynthOCT_Baseline/tree/main/Metrics_evaluation and consistency test https://github.com/SynthOCTChallenge/SynthOCT_Baseline/tree/main/self-consistency_check), we have established the following expectations for the estimates:

Structural MS-SSIM: Successful convergence is expected to yield a median SSIM > 0.3. Raw structural OCT expected moderate MS-SSIM values due to speckle patterns on raw structural OCT scans that are not additive noise — they result from optical interference, are therefore multiplicative, and have 100% magnitude modulation, which are hard to replicate exactly at the pixel level (therefore, we utilize speckle pattern statistics—physics-based texture patterns). Values significantly below 0.3 indicate a failure in synthetic scan generation (please check MS-SSIM red bars here: https://github.com/SynthOCTChallenge/SynthOCT_Baseline/blob/main/Metrics_evaluation/Results_compressed.png).

Parametric Maps: We expect higher precision for Speckle Contrast (SC) and RSC maps (MS-SSIM > 0.50 – preliminary tests). Conversely, the Optical Attenuation Coefficient (OAC) is expected to show the lowest values of LPIPS (in our tests LPIPS < 0.30 – preliminary tests), while for SC and RSC the LPIPS have widest intervals (please check the details here: https://github.com/SynthOCTChallenge/SynthOCT_Baseline/blob/main/Metrics_evaluation/Results_compressed.png).

Provide a description of how variability of the performance of individual algorithms across tests cases is assessed (e.g. SD across test cases, IQR, graphs, reporting outliers...).

For this specific challenge task, intra-scan variability (stability of reconstruction within a single B-scan) is considered more critical than inter-case variability.

Consequently, our assessment strategy is as follows:

Intra-Scan Analysis: We compute the MS-SSIM map for each scan and analyze the 95% interval of the local MS-SSIM values. This accounts for the fact that synthetic scans may contain localized regions with significant artifacts (outliers).

Robust Aggregation: To mitigate the impact of these local outliers, we utilize the median SSIM (rather than the mean) to derive a single representative score for each test case.

Across-Case Variability: The variability of these median scores across the 60 test cases will be reported.

Provide a description of how variability of rankings is assessed.

We will assess the stability of the rankings by analyzing the statistical significance of the differences between the participants' median SSIM scores. This assessment relies on the 95% Confidence Intervals (CIs) of the median estimates calculated across the 60 final test scans. Overlapping confidence intervals between ranked algorithms will indicate that the specific ranking order is subject to variability and that the performance difference is not statistically distinct.

Provide a description of statistical tests that are used to assess whether the differences in performance between algorithms are statistically significant.

To formally assess whether the performance differences between algorithms are statistically significant, we will additionally employ the Wilcoxon signed-rank test.

This non-parametric paired test is selected because:

Paired Design: All algorithms are evaluated on the exact same set of 60 test cases.

Non-Normality: The distribution of MSSSIM scores (and their differences) may not follow a normal distribution, making parametric tests (like the t-test) less suitable.

Procedure: We will specifically test the top-ranked algorithms against each other. If the 95% Confidence Intervals of their median scores overlap, the Wilcoxon test will be calculated to determine if the null hypothesis (that the median difference between the algorithms is zero) can be rejected at a significance level of 5%.

Provide a description of the missing data handling.

We do not anticipate missing data, as the algorithms are expected to process the provided dataset uniformly. However, should missing results occur due to technical failures, they will be handled according to the following protocol:

Systemic Data Failure: If a specific test case fails to be processed by all participants (indicating a corruption or intrinsic issue with the provided scan itself), this case will be excluded from the final evaluation for all teams to ensure fairness.

Participant-Specific Failure: If a test case fails only for certain participants while others successfully process it, the missing result will be treated as an algorithmic failure. In such cases, the participant will receive a penalty (e.g., the lowest possible score for that case), reflecting the algorithm's lack of robustness compared to competing solutions that successfully handled the same input.

Indicate any software product that is used for all data analysis methods.

We utilize Python and its standard statistical packages for all data analysis methods.

Further analyses

Present further analyses to be performed (if applicable), e.g. related to

- combining algorithms via ensembling,
- inter-algorithm variability,
- common problems/biases of the submitted methods, or
- ranking variability.

We plan to conduct a comprehensive post-challenge analysis to identify the specific methodological factors driving performance. This includes:

Methodological Correlation: We will analyze the correlation between specific algorithmic choices and their success in specific metric groups.

Trade-off Analysis: Special attention will be paid to the trade-offs between visual fidelity (Structural MS-SSIM) and physical consistency (OAC/RSC convergence). We aim to identify which approaches successfully bridge the gap between realistic texture generation and accurate physical parameter modeling.

Comparison with SOTA: For the final summary paper, we plan to train GAN and Diffusion models to compare our physics-based, scatterer-driven approach with SOTA generative methods. Due to the non-pairwise nature of this comparison, we will evaluate the FID metric to benchmark the distributions of real scans, our synthetic output, and SOTA-generated scans based on the underlying physics-based parametric maps.

Dissemination: The results of these analyses will be consolidated into a joint challenge summary paper.

Contributed teams will be invited to co-author this publication (additionally to their proceedings) to provide in-depth descriptions of their solutions and contribute to the discussion on the future of digital phantoms in OCT.

ADDITIONAL POINTS

References

Please include any reference important for the challenge design, for example publications on the data, the annotation process or the chosen metrics as well as DOIs referring to data or code.

Dataset: <https://zenodo.org/records/18095266> (DOI: 10.5281/zenodo.18095266)

Code repo: <https://github.com/SynthOCTChallenge> (including evaluation codes)

Previous Related Cup: <https://2024.ieeeicip.org/vip-cup/>

This Challenge platform: <http://synthoct.com/>

Papers, related to this Challenge:

[1] Sovetsky, A., Matveyev, A., Chizhov, P., Zaitsev, V., Matveev, L. (2025). OCT Scans Simulation Framework for Data Augmentation and Controlled Evaluation of Signal Processing Approaches. In: Fernandez, V., Wolterink, J.M., Wiesner, D., Remedios, S., Zuo, L., Casamitjana, A. (eds) Simulation and Synthesis in Medical Imaging. SASHIMI 2024. Lecture Notes in Computer Science, vol 15187. Springer, Cham.

https://doi.org/10.1007/978-3-031-73281-2_12

[2] Nikoshin, D., Mikhailenko, D., Sovetsky, A., Matveyev, A., Zaitsev, V., Matveev, L. (2026). From Tissue-Mimicking Phantoms to Physics-Based Scans: Synthetic OCT for Few-Shot Foundation Model Training. In: Fernandez, V., Wiesner, D., Zuo, L., Casamitjana, A., Remedios, S.W. (eds) Simulation and Synthesis in Medical Imaging. SASHIMI 2025. Lecture Notes in Computer Science, vol 16085. Springer, Cham.

https://doi.org/10.1007/978-3-032-05573-6_5

[3] Petrova GA, Petrova KS, Nemirova SV, Karpenko AA. 3D optical coherence tomography: possibilities in the diagnosing skin neoplasms. Russian Journal of Clinical Dermatology and Venereology. 2019;18(6):715727. (In

Russ.)

<https://doi.org/10.17116/klinderma201918061715>

[4] A. A. Sovetsky, K. S. Petrova, M. A. Brueva, M. G. Ryabkov, A. L. Matveyev, L. A. Matveev, V. Y. Zaitsev.

"Automated segmentation and skin-layer thickness estimation by extracting the optical scattering coefficient and speckle contrast parameter from optical coherence tomography scans". Skin Pharmacology and Physiology, 2026

Further comments

Further comments from the organizers.

This challenge introduces a pioneering paradigm: the generation of physics-based digital phantoms rather than simple image synthesis. This approach has far-reaching implications beyond the immediate scope of the challenge:

Cross-Vendor Standardization: By enabling virtual scanning simulations where device parameters can be adjusted in silico, these phantoms allow for the generation of synthetic data that mimics different hardware vendors, facilitating robust cross-device generalization.

Perfect Ground Truth: This method solves the "annotation bottleneck" by creating synthetic scans with ideal, mathematically defined ground truth, which is impossible to obtain with in vivo imaging.

Targeted Augmentation: It enables the creation of datasets with specific feature representations, allowing researchers to generate "corner cases" or rare pathologies to focus model training on specific biological features.

Virtual Histology: It lays the foundation for accurate "optical biopsy" and "virtual histology" by reconstructing the underlying physical properties of tissue (scatterer distributions) rather than just its visual texture. We emphasize that a key goal of this challenge is to advance the development of non-invasive OCT-based virtual histology. This involves first solving the inverse problem: mapping the underlying scatterer distributions and their parameters within target real OCT scans.

Challenge License Statement (as per the request of the MICCAI 2026 Challenge Team): We confirm that we use the CC BY 4.0 license. The data is hosted on Zenodo (<https://zenodo.org/records/18095266>) providing open access without registration barriers, which is fully consistent with the License terms and the guidelines provided by The MICCAI 2026 Challenge Team.